

Eco-Loop — Building Agent Savings Report

EnergyPlus 5-zone office · Chicago TMY3 · 15-minute control timestep

Run period 7/1 – 7/21

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OVERVIEW

Key metrics

Identical building, weather and run period across all three controllers — the only variable is the logic making the decisions.

FACILITY
ELECTRICITY SAVED

3.09%

2965 kWh vs 3060 kWh baseline

HVAC ELECTRICITY
SAVED

10.7%

the portion setpoints actually control

COMFORT
VIOLATIONS

0.00%

occupied zone-hours outside 20.0–25.0 °C

BEATS RULE-BASED
BY

+0.82 pp

rule-based saved 2.27%

MODEL

qwen2.5:3b-instruct

DECISIONS

136

MEDIAN LATENCY

3.3 s

SELF-CORRECTIONS

10

INVALID JSON

0

FALLBACKS

0

CACHE HIT RATE

31.0%

DATA

Results at a glance

Explicit numbers behind every chart on this page, read straight from eplusout.csv — the EnergyPlus output itself.

Controller comparison

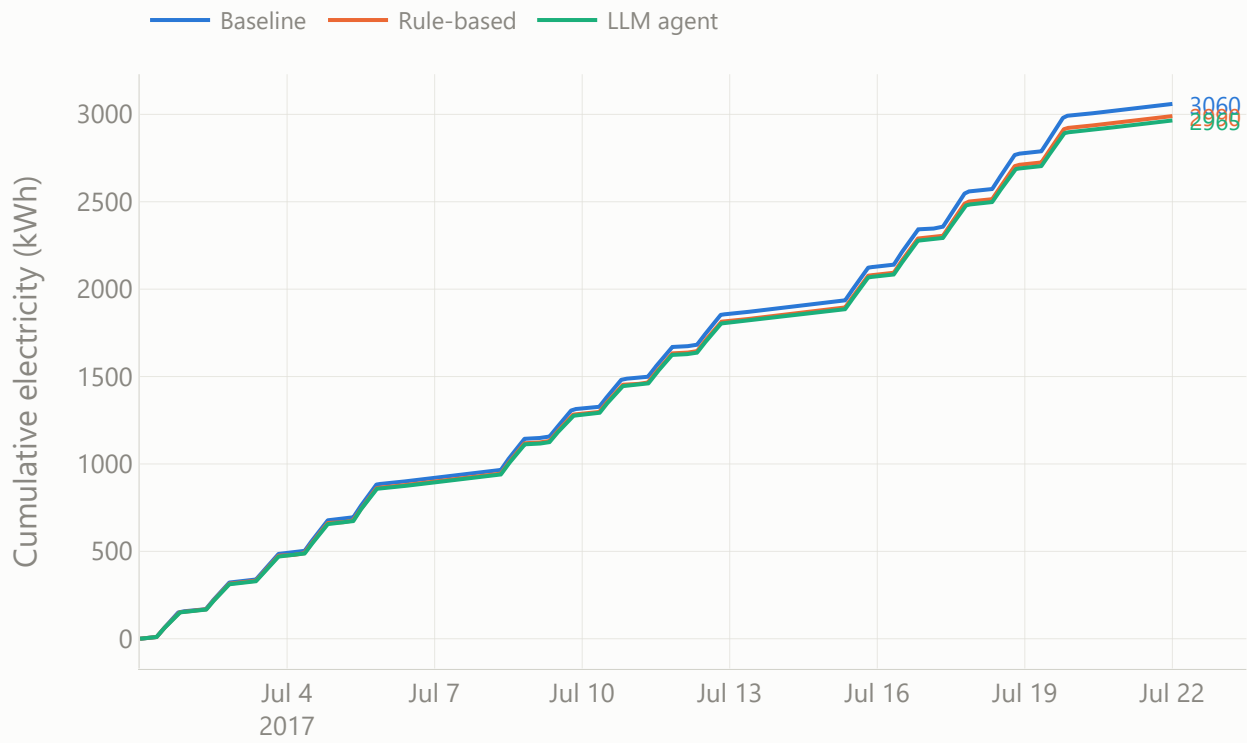
MODE	KWH	SAVED	HVAC KWH	PEAK KW	COMFORT VIOLATIONS
Baseline (stock schedules)	3059.74	—	904.23	20.22	0.00%
Rule-based (no LLM)	2990.15	+2.27%	832.82	19.65	0.00%
LLM agent (supervisory)	2965.24	+3.09%	807.45	19.59	0.00%

Trends over the run

How each controller's electricity use and comfort track over time.

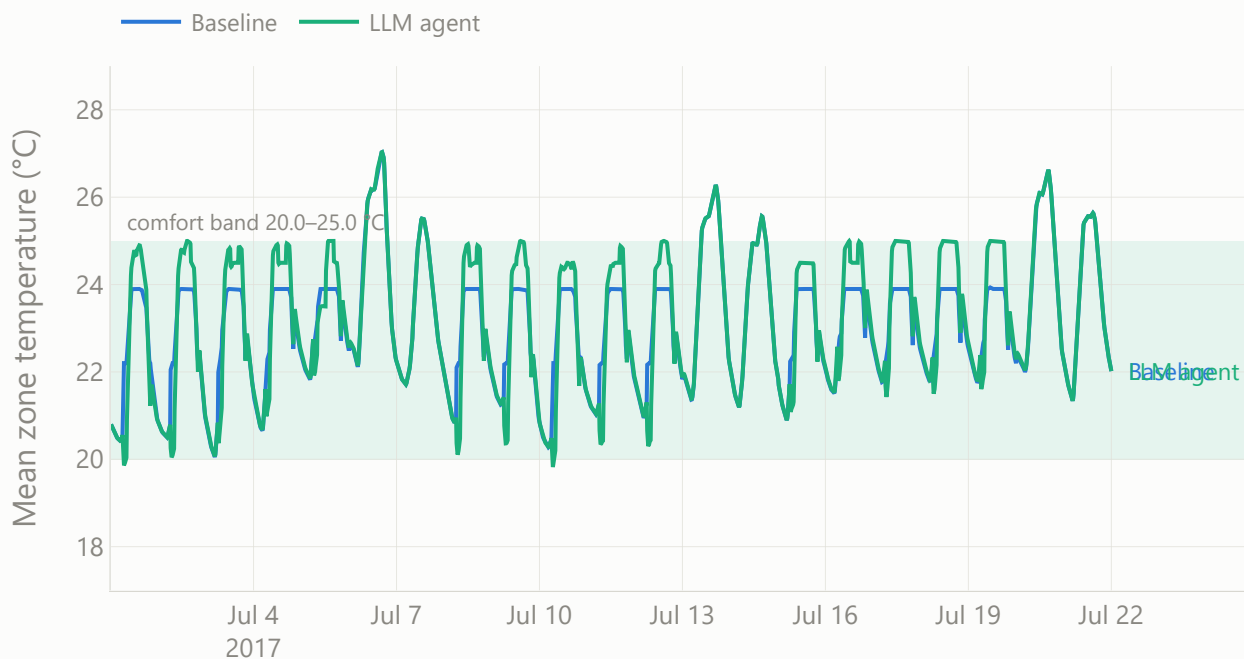
Cumulative electricity

The two controlled runs diverge from baseline as the run goes on. Lower is better.



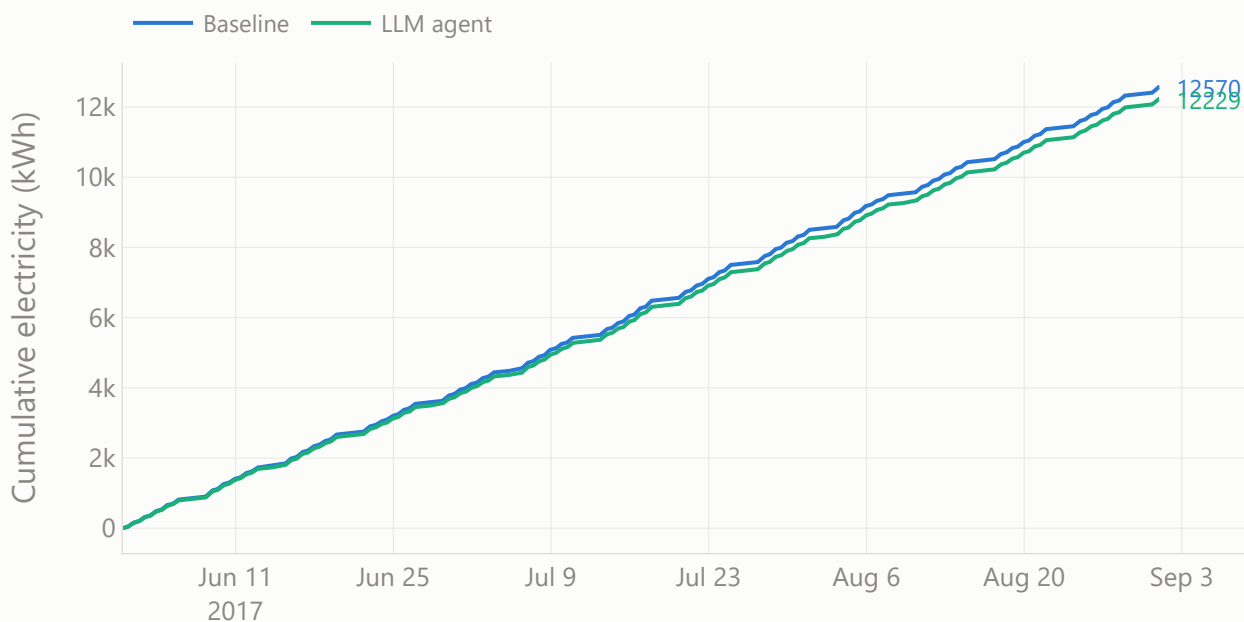
Comfort was maintained

Mean zone temperature against the 20.0–25.0 °C comfort band.



Endurance: a full cooling season

The identical loop run unattended from 1 June to 31 August — 8,832 control timesteps, 4.4× the reported experiment.



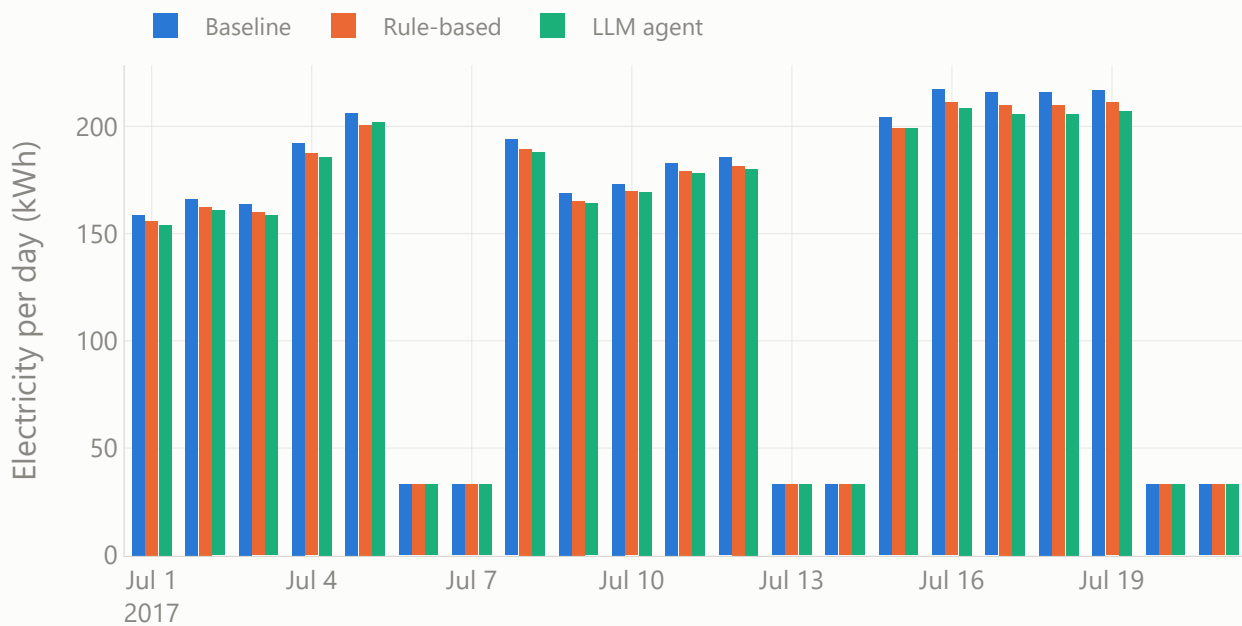
BAR CHARTS

Comparing the controllers

Same run, same building — only the controller differs.

Electricity per day

Savings concentrate on hot weekdays, when cooling actually runs.

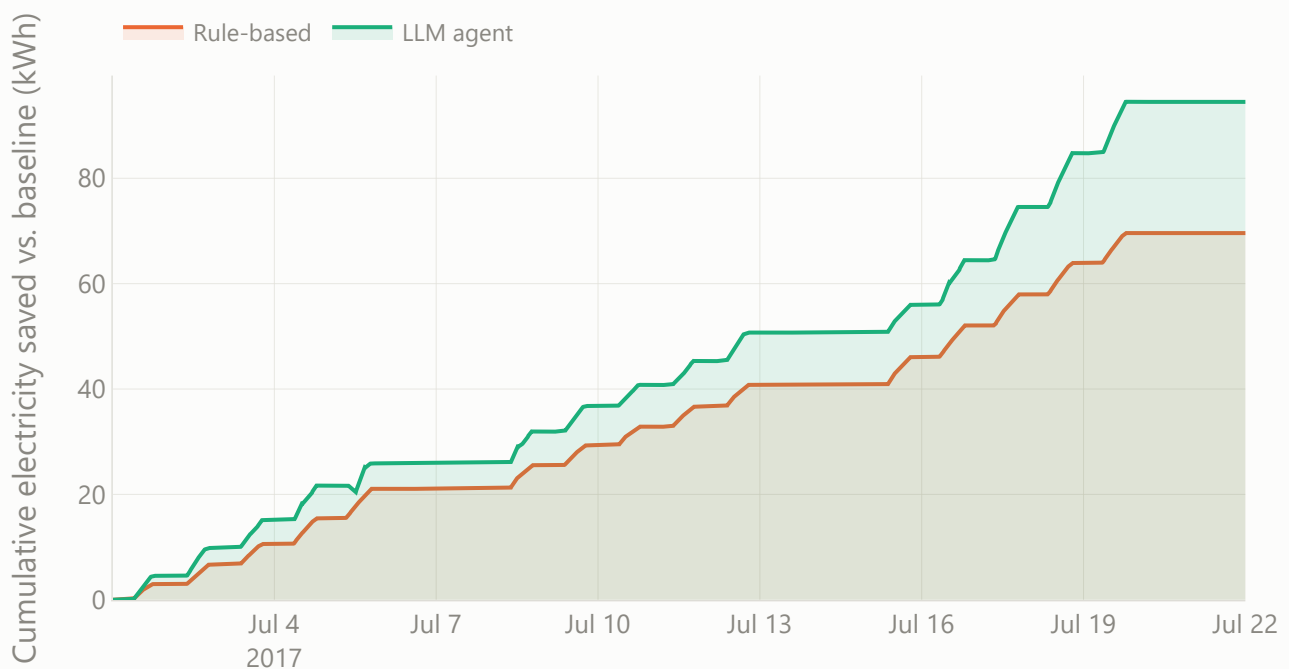


AREA CHART

Savings accumulating over time

Electricity saved, accumulating over the run

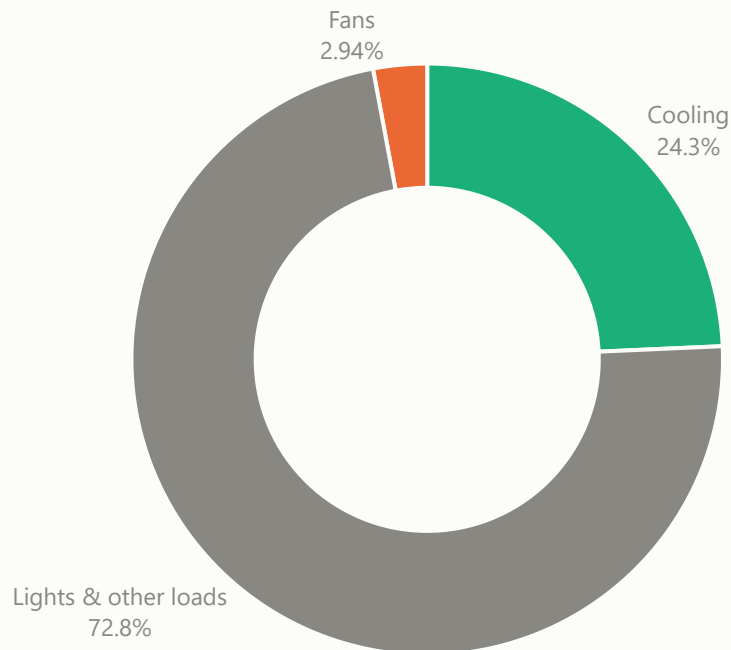
Baseline's cumulative use minus each controller's, at every timestep. The gap between the two lines is the 1.05-point contribution of the LLM's situational reasoning over a fixed rule.



Where the electricity goes

Where the electricity goes

Cooling and fans are what a thermostat can influence; everything else is out of reach — the honest reason the HVAC-only savings figure is reported alongside the facility-wide one.



ROBUSTNESS

Does it hold up over a long run?

The reported experiment is three weeks. This is the identical pipeline over three months. Nothing that must stay at zero moved, and the decision cache became markedly more effective.

3-week run vs. season-long endurance run

MEASURE	REPORTED RUN	ENDURANCE RUN
Simulated period	1–21 July (3 weeks)	1 June – 31 August (3 months)
Control timesteps	2,016	8,832
Supervisory decisions	126	552
Ran to completion	yes	yes
Crashes / hangs	0	0
Safety-clamp violations	0	0
Comfort violations	0.00%	0.00%
Malformed JSON	0	0
Fell back to rule-based	0	5
Decision cache hit rate	31.0%	59.8%
Electricity saved	+3.09%	+2.72%